
Función corriente:

Coordenadas cartesianas

$$v_x = \frac{\partial \psi}{\partial y}, \quad v_y = -\frac{\partial \psi}{\partial x}$$

Coordenadas cilíndricas

$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \phi}, \quad v_\phi = -\frac{\partial \psi}{\partial r}$$

Coordenadas esféricas

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \psi}{\partial r}$$

Operaciones en coordenadas esféricas:

Divergencia de $\vec{u}$

La divergencia en coordenadas esféricas está dada por:

$$\nabla \cdot \vec{u} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta u_\theta) + \frac{1}{r \sin \theta} \frac{\partial u_\phi}{\partial \phi}$$

Rotor de $\vec{u}$

$$\nabla \times \mathbf{u} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ u_r & r u_\theta & r \sin \theta u_\phi \end{vmatrix}$$

Laplaciano

Sea $\vec{A} = A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi}$ un campo vectorial que depende de las coordenadas esféricas (r, θ, ϕ) . Su laplaciana está dada por:

$$\begin{aligned} \nabla^2 \vec{A} = & \left[\nabla^2 A_r - \frac{2A_r}{r^2} - \frac{2}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) - \frac{2}{r^2 \sin \theta} \frac{\partial A_\phi}{\partial \phi} \right] \hat{r} + \\ & \left[\nabla^2 A_\theta - \frac{A_\theta}{r^2 \sin^2 \theta} + \frac{2}{r^2} \frac{\partial A_r}{\partial \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial A_\phi}{\partial \phi} \right] \hat{\theta} + \\ & \left[\nabla^2 A_\phi - \frac{A_\phi}{r^2 \sin^2 \theta} + \frac{2}{r^2 \sin \theta} \frac{\partial A_r}{\partial \phi} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial A_\theta}{\partial \phi} \right] \hat{\phi}. \end{aligned}$$

Por otro lado, la laplaciana de un escalar A está dada por:

$$\nabla^2 A = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial A}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial A}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 A}{\partial \phi^2}.$$

Operaciones en coordenadas cilíndricas:

Divergencia de $\vec{u}$

La divergencia en coordenadas cilíndricas está dada por:

$$\nabla \cdot \vec{u} = \frac{1}{r} \frac{\partial}{\partial r} (ru_r) + \frac{1}{r} \frac{\partial u_\phi}{\partial \phi} + \frac{\partial u_z}{\partial z}$$

Rotor de $\vec{u}$

$$\nabla \times \mathbf{u} = \frac{1}{r} \begin{vmatrix} \hat{r} & \hat{\phi} & \hat{z} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ u_r & ru_\phi & u_z \end{vmatrix}$$

Resolviendo el determinante, el rotor es:

$$= \hat{r} \left(\frac{\partial u_z}{\partial \phi} - \frac{\partial(ru_\phi)}{\partial z} \right) + \hat{\phi} \left(\frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r} \right) + \hat{z} \frac{1}{r} \left(\frac{\partial(ru_\phi)}{\partial r} - \frac{\partial u_r}{\partial \phi} \right)$$

Laplaciano

Sea $\vec{A} = A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z}$ un campo vectorial en coordenadas cilíndricas (ρ, ϕ, z) . Su laplaciana está dada por:

$$\begin{aligned} \nabla^2 \vec{A} = & \left[\nabla^2 A_\rho - \frac{2A_\rho}{\rho^2} - \frac{2}{\rho^2} \frac{\partial A_\phi}{\partial \phi} \right] \hat{\rho} + \\ & \left[\nabla^2 A_\phi - \frac{A_\phi}{\rho^2} + \frac{2}{\rho^2} \frac{\partial A_\rho}{\partial \phi} \right] \hat{\phi} + \\ & [\nabla^2 A_z] \hat{z}. \end{aligned}$$

Por otro lado, la laplaciana de un escalar A está dada por:

$$\nabla^2 A = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial A}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 A}{\partial \phi^2} + \frac{\partial^2 A}{\partial z^2}.$$

Navier–Stokes

$$\boxed{\begin{aligned}\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) &= -\nabla p + \mu \nabla^2 \mathbf{v} + \mathbf{f} \\ \nabla \cdot \mathbf{v} &= 0\end{aligned}}$$

Coordenadas Cartesianas

Componente x:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x$$

Componente y:

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho g_y$$

Componente z:

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g_z$$

La ecuación de continuidad es:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Coordenadas Cilíndricas

Componente r:

$$\begin{aligned}\rho \left(\frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_r}{\partial \theta} + u_z \frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r} \right) = \\ -\frac{\partial p}{\partial r} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_r}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial^2 u_r}{\partial z^2} - \frac{u_r}{r^2} \right] + \rho g_r\end{aligned}$$

Componente θ :

$$\begin{aligned}\rho \left(\frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_\theta}{\partial \theta} + u_z \frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} \right) = \\ -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_\theta}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{\partial^2 u_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2} \right] + \rho g_\theta\end{aligned}$$

Componente z:

$$\rho \left(\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_z}{\partial \theta} + u_z \frac{\partial u_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2} \right] + \rho g_z$$

La ecuación de continuidad es:

$$\frac{1}{r} \frac{\partial}{\partial r} (ru_r) + \frac{\partial u_z}{\partial z} = 0$$

Para un flujo axisimétrico: $u_\theta = 0$

Coordenadas Esféricas:

Componente r:

$$\begin{aligned} \rho \left(\frac{\partial u_r}{\partial t} + u_r \frac{\partial u_r}{\partial r} + \frac{u_\phi}{r \sin(\theta)} \frac{\partial u_r}{\partial \phi} + \frac{u_\theta}{r} \frac{\partial u_r}{\partial \theta} - \frac{u_\phi^2 + u_\theta^2}{r} \right) &= -\frac{\partial p}{\partial r} \\ \rho g_r \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u_r}{\partial r} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 u_r}{\partial \phi^2} + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial u_r}{\partial \theta} \right) \right] &- \\ + \rho g_r \mu \left[2 \frac{u_r + \frac{\partial u_\theta}{\partial \theta} + u_\theta \cot(\theta)}{r^2} + \frac{2}{r^2 \sin(\theta)} \frac{\partial u_\phi}{\partial \phi} \right] & \end{aligned}$$

Componente θ :

$$\begin{aligned} \rho \left(\frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\phi}{r \sin(\theta)} \frac{\partial u_\theta}{\partial \phi} + \frac{u_\theta}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r u_\theta - u_\phi^2 \cot(\theta)}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} \\ + \rho g_\theta \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u_\theta}{\partial r} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 u_\theta}{\partial \phi^2} + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial u_\theta}{\partial \theta} \right) + \frac{2}{r^2} \frac{\partial u_r}{\partial \theta} - \frac{u_\theta + 2 \cos(\theta) \frac{\partial u_\phi}{\partial \phi}}{r^2 \sin^2(\theta)} \right] & \end{aligned}$$

Componente ϕ :

$$\begin{aligned} \rho \left(\frac{\partial u_\phi}{\partial t} + u_r \frac{\partial u_\phi}{\partial r} + \frac{u_\phi}{r \sin(\theta)} \frac{\partial u_\phi}{\partial \phi} + \frac{u_\theta}{r} \frac{\partial u_\phi}{\partial \theta} + \frac{u_r u_\phi + u_\phi u_\theta \cot(\theta)}{r} \right) &= -\frac{1}{r \sin(\theta)} \frac{\partial p}{\partial \phi} \\ + \rho g_\phi \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u_\phi}{\partial r} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 u_\phi}{\partial \phi^2} + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial u_\phi}{\partial \theta} \right) + \frac{2 \frac{\partial u_r}{\partial \phi} + 2 \cos(\theta) \frac{\partial u_\theta}{\partial \phi} - u_\phi}{r^2 \sin^2(\theta)} \right] & \end{aligned}$$

La ecuación de continuidad es:

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_r) + \frac{1}{r \sin(\theta)} \frac{\partial u_\phi}{\partial \phi} + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} (\sin(\theta) u_\theta) = 0$$